

Session Objectives

- Describe the three primary energy systems that fuel activity./Learning Objective 1
- Explain how VO_2 , METs, and RER relate to exercise intensity fuel utilization and weight loss./Learning Objective 2.
- Describe the cardiovascular and respiratory responses to exercise and explain how these responses guide exercise prescription in individuals with chronic disease./Learning Objective 3
- Understand relative and absolute contraindications to exercise testing ./ Learning Objective 4.
- Describe how ventilatory and cardiac output responses are influenced by various physiological and environmental factors, including exercise intensity, temperature, and altitude./Learning Objective 5

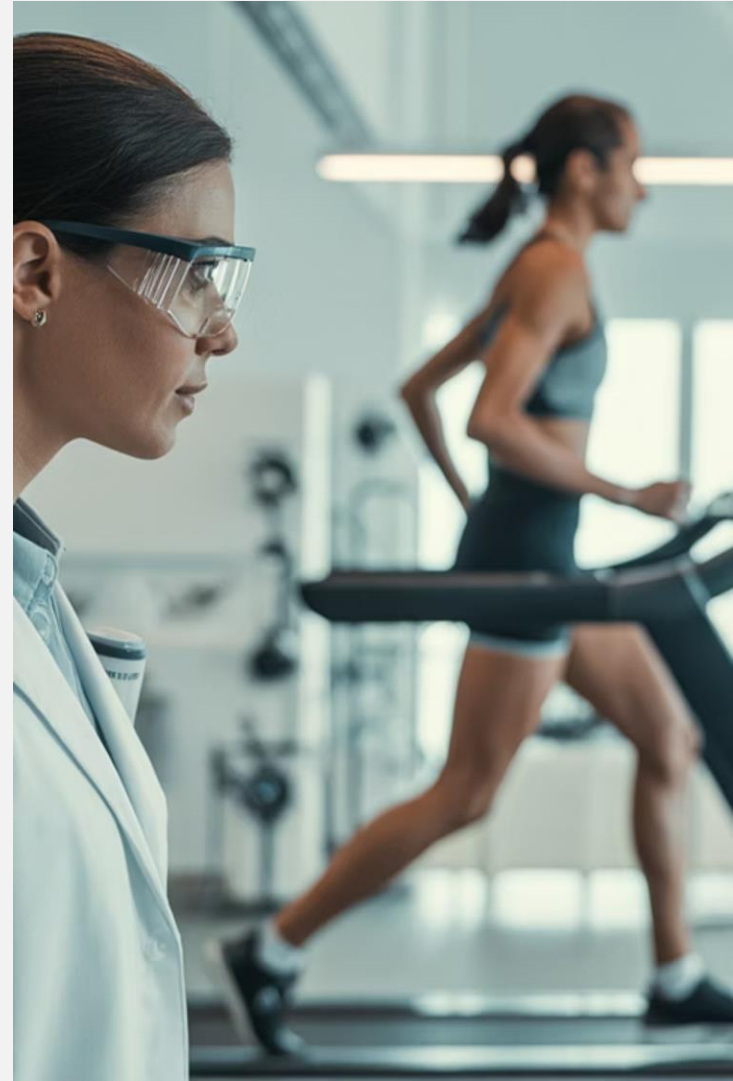
Lecture Outline: Intro to Exercise Physiology

Topic	Time (min)
Introduction to Exercise Physiology (definition, history)	4
Energy Systems (Phosphagen, Glycolysis, Oxidative)	7
VO ₂ Max, METs, and Measuring Intensity	6
Cardiovascular and Pulmonary Adaptations to Exercise	7
Muscle Fiber Types & Adaptation (Slow vs. Fast Twitch)	5
Exercise Prescription Principles (FITT-VP, intensity, ACSM)	7
Clinical Case: Fat Loss, Caloric Burn, RER, and Application	6
Me rambling	3

What Is Exercise Physiology?

From walking to resistance training, exercise physiology helps us understand how everyday movement impacts our bodies at the cellular, tissue, and systemic levels.

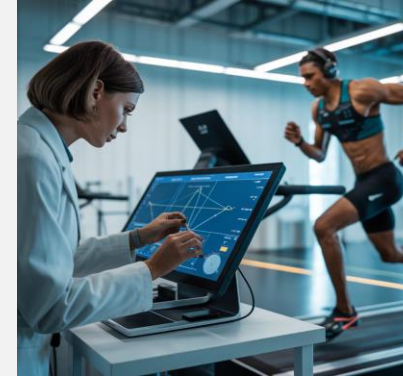
(4 slides)



Historical Background and Scope

Sport Exercise Physiology

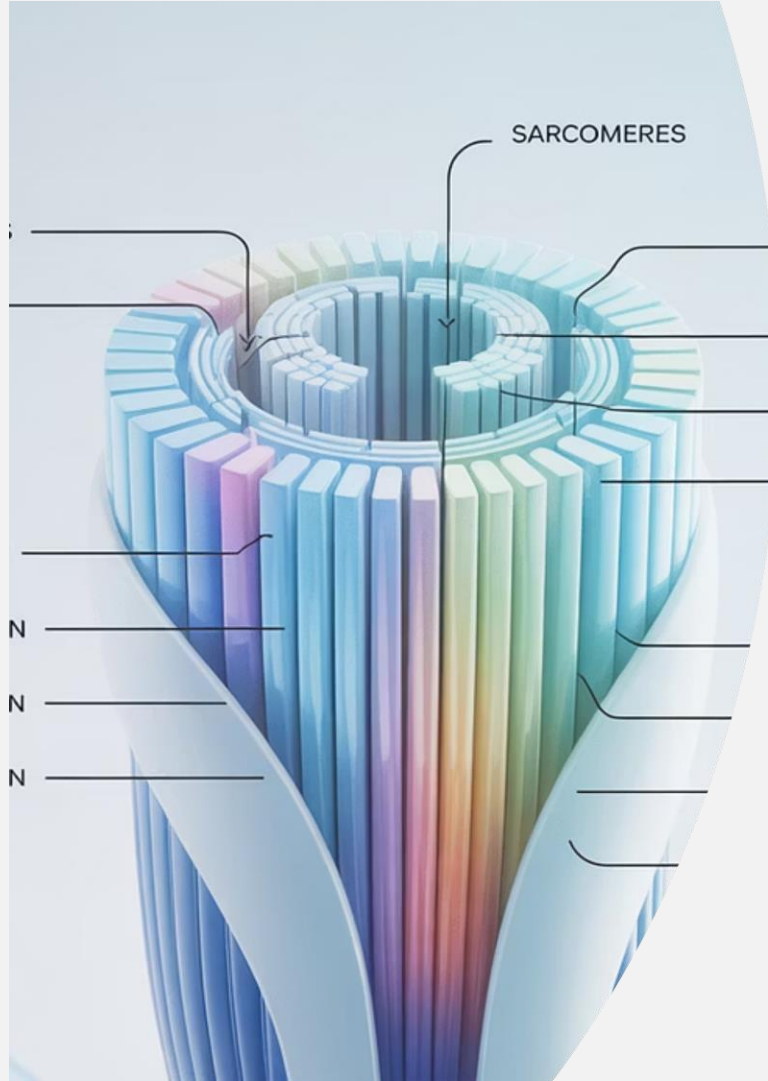
Focuses on optimizing athletic performance and training methodologies for competitive athletes and sports teams



Clinical Exercise Physiology

Emphasizes therapeutic applications for disease management, rehabilitation, and improving quality of life in clinical populations





Physiological Systems Involved

1

Musculoskeletal System

Powers movement through sarcomeres
microscopic contractile units containing actin and myosin
filaments that slide past each other during muscle
contraction, generating force and movement

2

Cardiovascular System

Increases heart rate
stroke volume
cardiac output
delivering oxygen and nutrients to working muscles while
removing metabolic waste products

3

Respiratory System

Enhances oxygen uptake (VO_2) and carbon dioxide
elimination through increased ventilation rate and depth,
maintaining blood gas homeostasis during exertion

4

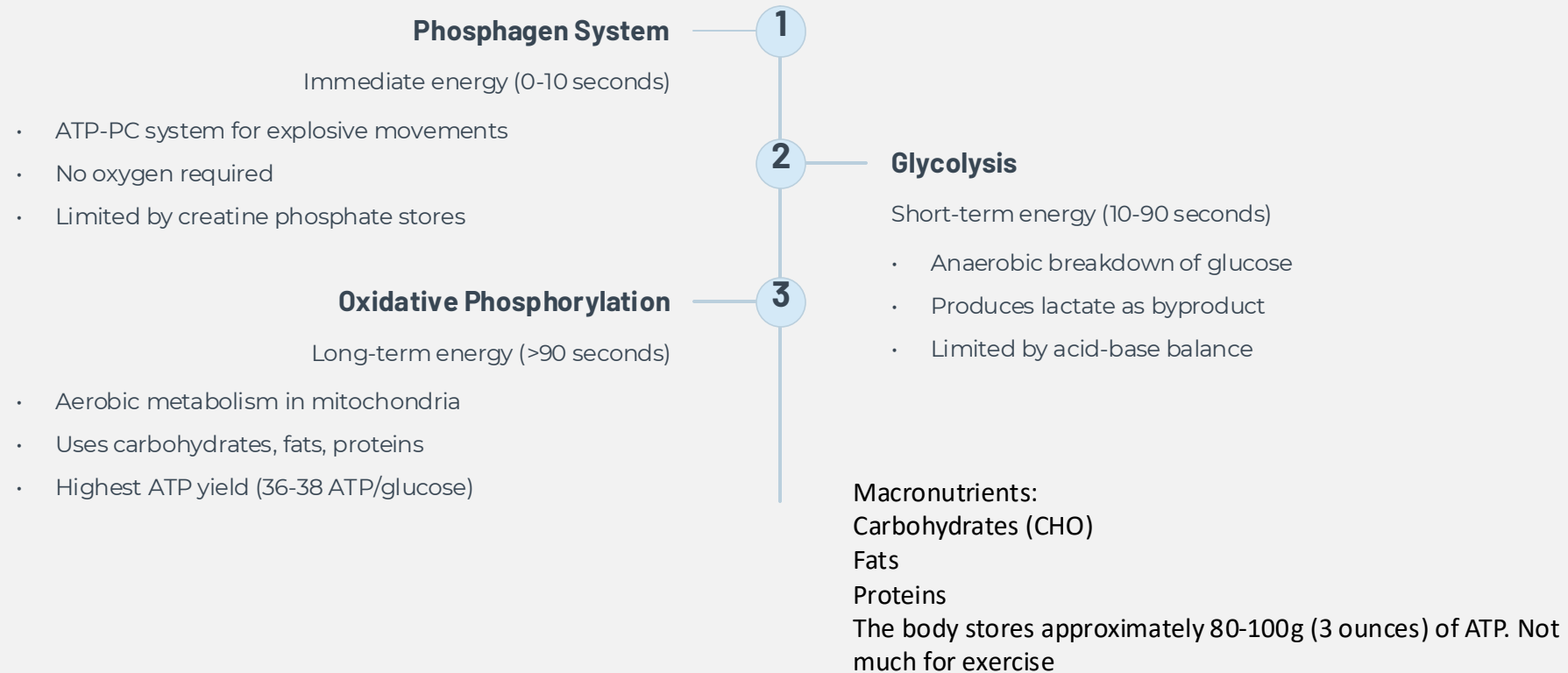
Nervous System

Coordinates movement patterns and activates sympathetic
responses that increase alertness, heart rate, and blood flow to
muscles while inhibiting digestive processes



Energy Systems & Metabolism

How muscles acquire and utilize ATP



📌 **Clinical note:** Statin-induced myopathy may result from impaired mitochondrial function, directly affecting oxidative phosphorylation and reducing exercise tolerance in patients.

Activity and Primary System Used

Duration of activity	Intensity of activity	Primary energy system
0-10 seconds	Extremely high	Phosphagen
10-30 seconds	Very high	Phosphagen and fast glycolysis
30 sec-2 min	High	Fast glycolysis
2-3 minutes	Moderate	Fast glycolysis and oxidative system
> 3 minutes	Low	Oxidative system

Indirect Calorimetry

- Calorie- The energy needed to raise 1 gram of H₂O 1 degree Celsius

Indirect calorimetry Measured by $\dot{V}_E$ and exchange of CO₂ and O₂

- Pure carbohydrate catabolism 5.05 Kcal/L O₂
- Fat catabolism 4.73 Kcal/Liter O₂
- the difference in caloric expenditure between pure carbohydrate and fat catabolism, of an average healthy person exercising for 30 min at a VO₂ (oxygen consumption) of 1.5 L/min, would amount to
- $5.05 * 1.5 \text{ L/min} = 7.6 \text{ kcal/min} * 30 \text{ min} = 228 \text{ kcal CHO}$
- $4.73 * 1.5 \text{ L/min} = 7.1 \text{ kcal/min} * 30 \text{ min} = 212 \text{ kcals fat}$

Intensity	Calories Burned	% from Fat	Fat Calories
Low	200 kcal/hr	70%	140 kcal
Moderate	400 kcal/hr	50%	200 kcal

If you're trying to lose fat, caloric deficit matters more than fat as a % fuel. Moderate to high-intensity workouts can burn more total fat in less time.

What affects your $\dot{V}O_2$

Genetics and Training

- Muscle type
- Oxidative enzymes
- A- $\dot{V}O_2$ difference
- Cardiac Output (Q)

Altitude

- O_2 saturation

Mode of Exercise

- Quantity of muscle activated
- TDM or Bike?
- Swimmer on Treadmill?

Gender

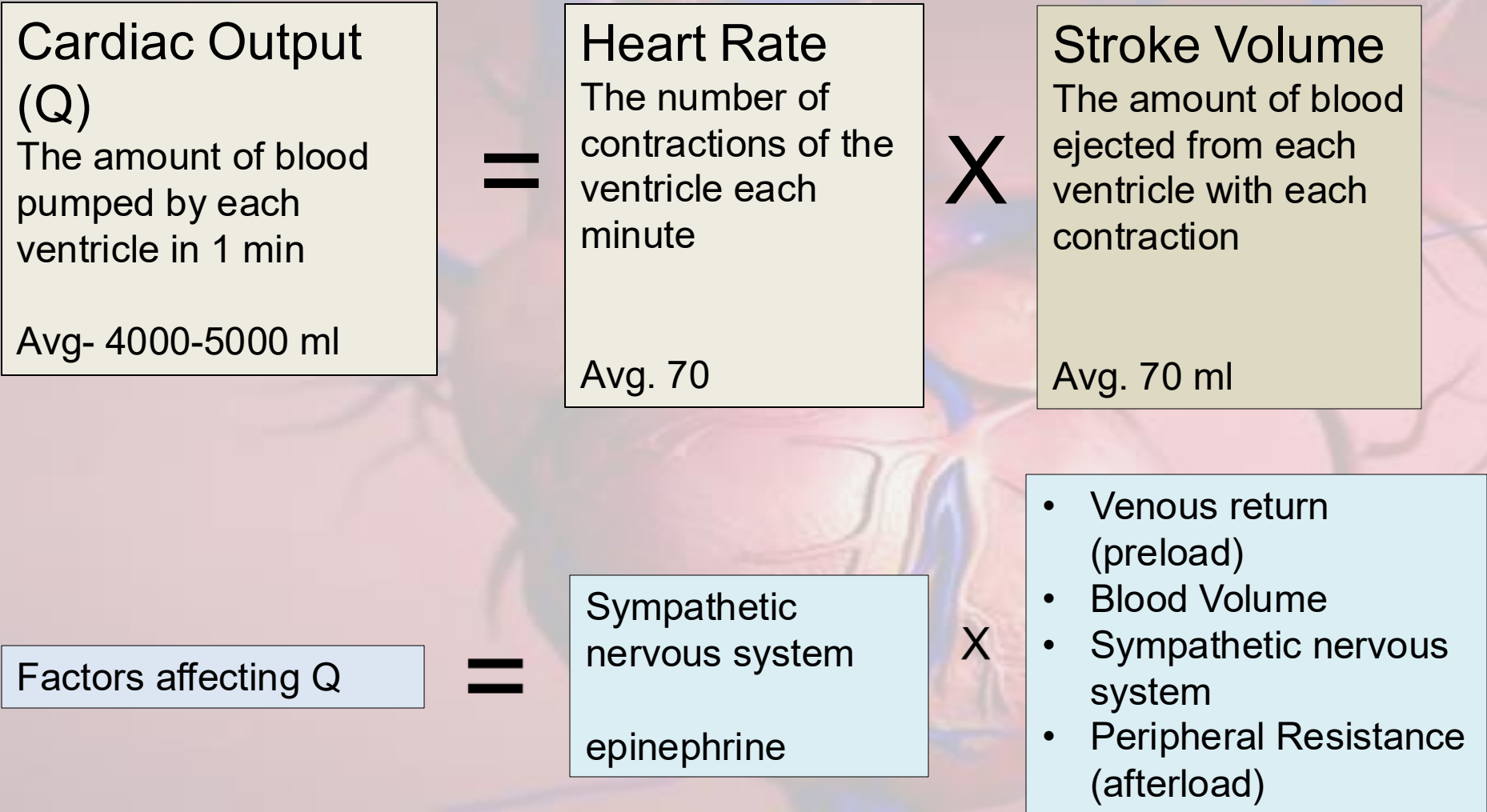
- Women tend to be lower
- Body composition
- Hemoglobin- allows for oxygen carrying ability

Age

- ~ 1% per year
- people who exercise maintain it
- loss can be greater for athletes due to initial value

*Q and a- $\dot{V}O_2$ difference are main factors in changing $\dot{V}O_2$.

Cardiac Output



Concentric vs Eccentric Ventricular Hypertrophy

Concentric hypertrophy- muscle of the left side of the pumping chamber increases in size and thickness. Chamber becomes smaller.

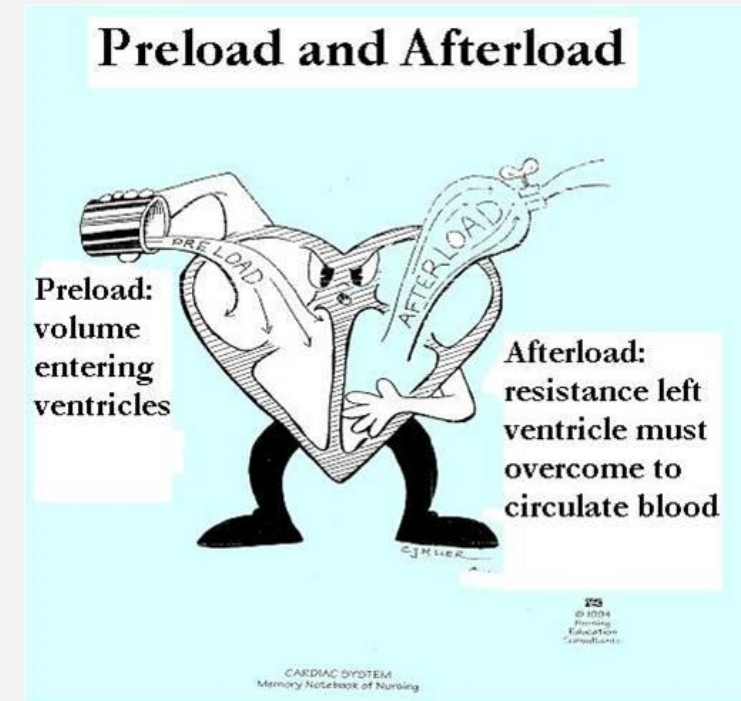
Pressure overload

↓ In stroke volume and cardiac output

- high blood pressure or any obstruction of outflow of blood from the left heart (e.g intense resistance training). Aortic valve narrowing.

Eccentric hypertrophy- muscle of the left side of the pumping chamber thickens, increases in size and dilates. Chamber becomes larger.

- chronic preload on the heart e.g. cardiovascular exercise.
- Volume overload



Medical Conditions That Decrease Stroke Volume

Condition	Mechanism
Heart Failure (HFrEF)	Reduced contractility, dilated ventricles
Myocardial Infarction	Damaged myocardium = impaired pump
Cardiac Tamponade	↓ Preload due to compression of the heart
Hypovolemia / Hemorrhage	↓ Preload from reduced blood volume
Pulmonary Embolism	Impaired RV output = ↓ LV preload
Severe Aortic Stenosis	↑ Afterload limits ejection
Tachycardia (severe)	↓ Filling time = ↓ Preload

Increases a-vO₂ Difference (↑ Oxygen Extraction):

Factor	Why
Exercise (especially endurance)	Muscles extract more O ₂ during increased demand
Anemia	↓ Hemoglobin → body compensates by extracting more O ₂
Sepsis (early stage)	↑ tissue metabolism → higher extraction
Heart failure (mild/moderate)	Poor perfusion = tissues compensate by extracting more
Peripheral arterial disease (PAD)	Poor delivery, so tissues extract more oxygen from what little arrives

Decreases a-vO₂ Difference (↓ Oxygen Extraction):

Factor	Why
Severe heart failure (HFrEF)	Impaired perfusion limits tissue extraction
Sepsis (late stage)	Mitochondrial dysfunction = poor oxygen utilization
Cyanide or mitochondrial toxins	Block cellular respiration → O ₂ not used despite being delivered
Shock (e.g., cardiogenic or distributive)	Tissues are underperfused and/or can't extract oxygen properly
AV fistulas or shunting	Bypasses tissue beds → less O ₂ extraction
Sedation or paralysis	↓ metabolic demand → less oxygen extracted

Heat and humidity are concerns for athletes competing outdoors at the Tokyo Olympics

High heat and humidity:

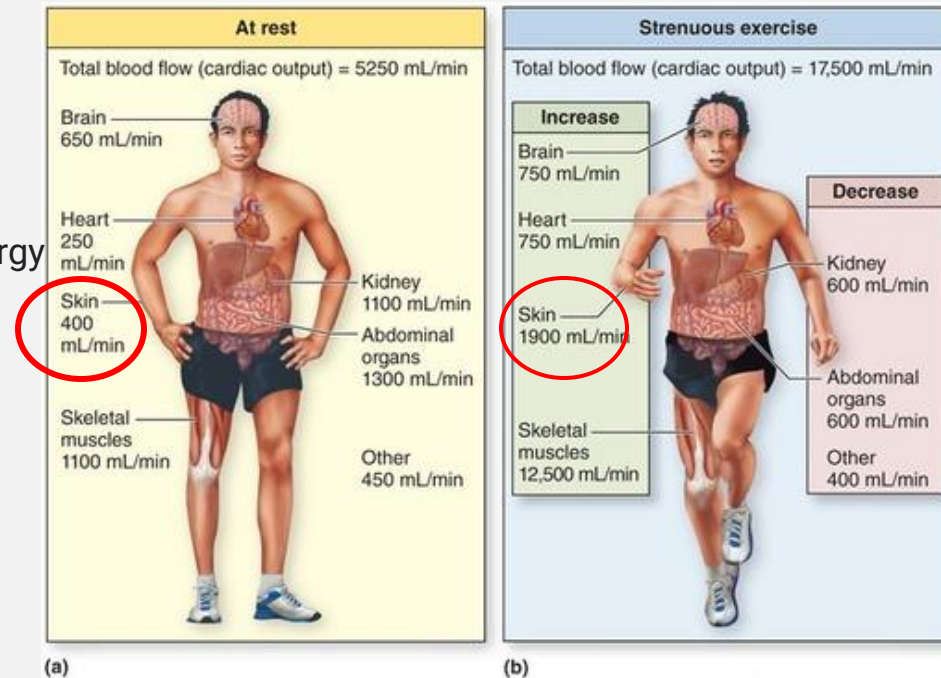
- 1) increased core body temperature
- 2) dehydration. ...

Causes decrease in VO₂max

body can't utilize oxygen as efficiently to provide energy



86.9 degrees Fahrenheit (30.5 degrees Celsius) high humidity



Why VO₂ max Decreases in Extreme Weather

Factor	Heat ☀️	Cold ❄️
Cardiac Output	Blood shunted to skin for cooling → ↓ central blood volume	Vasoconstriction ↑ afterload → ↑ cardiac strain
Muscle Oxygenation	Reduced muscle perfusion (due to skin blood flow priority)	Less oxygen delivery due to vasoconstriction
Oxygen Utilization	Enzyme efficiency drops if body overheats	Slowed metabolic reactions and nerve conduction
Hydration Status	Dehydration → ↓ plasma volume → ↓ stroke volume	Dehydration risk still present with cold diuresis
Pulmonary Function	↑ Breathing rate in heat → less efficient exchange	Cold, dry air → bronchoconstriction, discomfort
VO₂max Effect	↓ VO ₂ max due to cardiovascular and thermoregulatory strain	↓ VO ₂ max due to delivery and metabolic inefficiency

Altitude

Physiological changes occur >3,900 ft

Acute

- Reduced partial pressure of O₂
- Increased cardiac output (rest/exercise)
- Increase in HR
- SV remains the same
- Increase in V_e due to increase in BR
- Body fluids become more alkaline due to ↓ in CO₂ with hyperventilation

• Longer Term

- 10-14 days HR and Q begin to return to normal due to increased red blood cell production
- Increased capillarization
- Increased mitochondria
- Increased a-VO₂ diff
- Increased use of free fatty acids sparing muscle glycogen

Revert in 1 month

More dehydrated at altitude :

- You exhale and perspire (air humidity is lower)
- Can lose up to 1 quart or more of water per day than normal

For every 300 meters (1000 ft. about) above 1500 meters elevation (about 5000ft), VO₂max decreases by 1.5 to 3.5%.

E.g.

6000ft there is about 80% of the available oxygen you get at sea level.

Cardiovascular & Pulmonary Adaptations

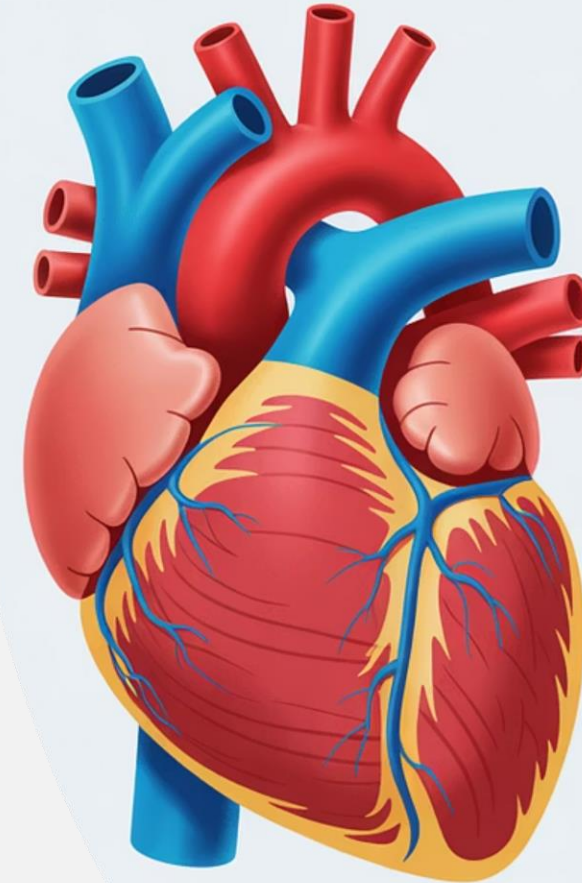
Acute Responses

- ↑ Heart rate (chronotropic)
- ↑ Stroke volume (inotropic)
- ↑ Cardiac output
- ↑ Systolic BP, ↓ Diastolic BP
- ↑ Respiratory rate and depth

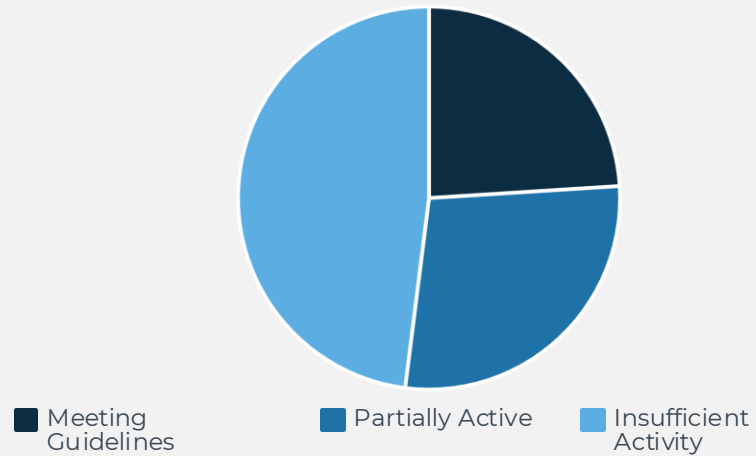
Chronic Adaptations

- ↓ Resting heart rate
- ↑ Blood volume and RBC mass
- ↑ Capillary density
- ↑ VO_2 max (15-20%)
- ↑ Ventilatory efficiency

⊗ **Clinical** Patients with Heart Failure with Preserved Ejection Fraction (HFpEF) fatigue quickly due to impaired diastolic filling and reduced stroke volume augmentation during exercise.



Challenges and Considerations



48% of adults worldwide do not meet recommended physical activity guidelines.

Healthcare providers face several barriers to effective exercise prescription, including time constraints during consultations, insufficient training in exercise science, and lack of reimbursement for counseling.

The most successful exercise interventions focus on enjoyable, sustainable activities that patients can realistically incorporate into their daily routines, with clear progression plans and regular follow-up.

Summary

1. The body uses three systems to generate ATP for movement: phosphagen, glycolytic, and oxidative.
2. Exercise intensity is the key factor that determines which energy system is dominant—not just duration or frequency.
3. VO_2 is influenced by multiple factors, including cardiac output, oxygen delivery, and muscle utilization.
4. Medical conditions (e.g., heart failure, COPD, neurological disorders), medications (e.g., beta blockers, vasodilators), and environmental stressors (e.g., heat, cold, altitude) can all alter normal physiological responses to exercise.

Clinical Scenario:

Patient:

32-year-old female, BMI 32, sedentary lifestyle

Complaint:

“I want to lose weight, mostly body fat. I’ve been told to keep my heart rate low to burn fat, but I don’t know if that’s enough.”

Vitals:

- Resting HR: 78 bpm
- BP: 118/76
- No known medical conditions
- Normal labs, no diabetes or thyroid issues

Current activity:

Walks 20 minutes 3x/week at a casual pace (approx. 50% max HR)

Based on her goal of fat loss and current fitness level, should you recommend she continue low-intensity walking or increase to moderate-to-high intensity exercise?

Use knowledge of energy systems, fat vs. carbohydrate utilization, and total caloric expenditure to support your recommendation.

Which of the following is the **most appropriate exercise recommendation** to support this patient's goal of **fat loss**?

- A.** Continue low-intensity walking to stay in the “fat-burning zone” since fat is used more at lower intensities.
- B.** Increase to moderate-intensity exercise (60–75% HRmax) to enhance total caloric expenditure and improve fitness.
- C.** Switch to high-intensity interval training (HIIT) only, since low-intensity exercise does not burn fat.
- D.** Avoid increasing intensity until weight loss begins, as moderate or high-intensity exercise could promote carbohydrate use and limit fat burning.

Case Integration: Clinical Application

"45-year-old sedentary male starts jogging, experiences rapid fatigue and lightheadedness"

1

Limiting Systems

- Underdeveloped oxidative energy system
- Poor cardiovascular conditioning (↓ stroke volume)
- Inefficient ventilatory response
- Low lactate threshold

2

Expected 4-Week Adaptations

- 10-15% increase in VO_2 max
- Improved stroke volume and cardiac output
- Enhanced mitochondrial density
- Increased lactate threshold
- Delayed onset of fatigue and symptoms

When prescribing exercise, remember that physiological adaptations occur across multiple systems simultaneously, creating a coordinated improvement in functional capacity.

FROM CPG Indications to exercise test

- Diagnose coronary artery disease
- Diagnose chest pain, shortness of breath or lightheadedness
- Determine a safe level of exercise
- Check the effectiveness of procedures done to improve coronary artery circulation in patients with coronary artery disease
- Predict risk of dangerous heart-related conditions such as a heart attack.
- Depending on the results of the exercise stress test, the physician may recommend more tests such as a nuclear stress test or cardiac catheterization

Indications for Exercise Testing Pulmonary Patients

- Cause of SOB remains unclear despite PFT's
- To distinguish whether cardiac or respiratory disease limits exercise tolerance
- To evaluate for exercise –induced O_2 desaturation
- To determine appropriate intensity for exercise training

Normal Exercise Responses

ACSM Guidelines 8th edition

Variable	Response	Clinical Significance
Heart Rate	Linear ↑ with increase intensity ~10 beats per MET	Reduced with Beta blockers, and certain medications
Systolic BP	Linear ↑ ~10 Hg/MET. May peak at max exercise	>250 mm Hg or hypotension > 10 mm Hg may indicate myocardial ischemia / LV dysfunction
Diastolic BP	No change or slight decrease	End test > 15 mm Hg > 115 mm Hg
V _O ₂	Linear ↑ with ↑ intensity	Will plateau and V _e will rise
O ₂ saturation	A ↓ of 4% is considered abnormal	May indicate respiratory disease
ST wave changes	There should be no upward or downward sloping	≥ 1.0 mm downslope Suggests myocardial ischemia ST- elevation may reflect MI aneurysm or wall motion abnormalities